
Rollout-Aware Coefficient Activity for Task Selection in Verifiable-Reward RL

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Abstract

RL with verifiable rewards often treats task curricula as independent of the group estimator that converts sampled outcomes into updates. We ask which tasks that estimator makes active. For the practical MaxRL convention that drops all-fail groups, the expected absolute coefficient mass over N i.i.d. binary rollouts is exactly $2(\text{pass}@N - \text{pass}@1)$, the estimator’s own $\text{pass}@N$ gap. Its peak moves toward harder tasks as N grows, it recovers $p(1 - p)$ at $N = 2$, and the all-fail rule shifts the corresponding truncation order to $N - 1$. This is an estimator-side diagnostic, not a theorem of learning progress. In a fresh 20-seed Acrobot tournament at $N = 16$, the deployed- N score improves target-uniform AUC over $p(1 - p)$ by $+0.0480$ (95% paired-bootstrap CI $[+.0209, +.0738]$) and also beats uniform sampling in this fixed eight-threshold family. An exact-probability Digits counter-test rejects a universal estimator-to-sampler mapping. On neural gradients, an externally recorded six-block maze factorial finds higher time-integrated coverage under MaxRL than GRPO in 6/6 blocks under each sampler, an estimator-conditioned ordering rather than a direct score test. Data-selection methods should therefore be evaluated with the estimator they ship, while coefficient activity is treated as a rollout-aware curriculum hypothesis rather than a universal objective.

1 Introduction

RL with verifiable rewards (RLVR) converts automatically checkable outcomes into policy updates, but every update first pays for a group of model rollouts. A task curriculum tries to make that budget productive by sampling tasks believed to be learnable now. Such curricula are often evaluated as portable modules, separately from the estimator that turns their sampled data into gradients.

That separation is fragile. Group-relative estimators emit no contrast when all N samples agree, while success-conditioned estimators may drop all-fail groups entirely. Variance normalization can continue weighting near-mastered prompts after a success-conditioned estimator has released them. A sampler and estimator can therefore move training mass toward different parts of the task pool even when both are described as improving efficiency.

We organize these interactions around the expected absolute coefficient mass emitted by the deployed estimator before accounting for score-gradient direction or magnitude. For practical MaxRL (Tajwar et al., 2026), this mass is

$$A_N(p) = 2\{1 - (1 - p)^N - p\} = 2(\text{pass}@N - \text{pass}@1).$$

Its two zeros separate a dead zone, a learnable region, and a mastered tail. Its peak moves toward harder prompts as the rollout count grows. Coefficient mass is not a proof of learning progress, but it is estimator-side activity a task sampler can predict before purchasing a group of rollouts.

One central claim — coefficient activity is a rollout-aware acquisition hypothesis derived from the deployed estimator — is supported by three rungs:

1. Derivation. We derive A_N , its peak and asymptotics, and show that the practical drop-all-fail estimator targets truncation order $T = N - 1$, not N ; a fixed-completion sweep checks its nontrivial N dependence. The identity is an estimator-side diagnostic, not a learning theorem.
2. Controlled scope test. In a fixed eight-threshold Acrobot pool, u_{16} scoring is supported over both $p(1 - p) = u_2$ scoring and uniform sampling by the frozen rule; all arms still deploy the same $N = 16$ practical-MaxRL estimator. In an exact-probability Digits counter-test, the internally frozen interaction fails, mismatched RLOO prefers the same sampler, and neither matched sampler beats uniform. Together these studies locate the supported boundary at rollout-aware difficulty targeting rather than a universal estimator-to-sampler mapping.
3. Supporting neural measurement. At common optimizer settings, the externally recorded fresh maze wave has higher MaxRL than GRPO coverage in all six blocks under each sampler. This tests an estimator-conditioned ordering, not FrontierMax itself. We preserve the external-record and missing-checkpoint limitations and recommend retaining raw task outcomes so mean@ k and standard pass@ k can be reported under the exact deployed estimator.

The paper is deliberately narrower than the full project record: exploratory mechanisms, failed hypotheses, robotics pilots, gate studies, and implementation detail remain in the appendix and artifact.

2 Coefficient activity under practical MaxRL

Setup. For task x , let $r_i \in \{0, 1\}$ be N conditionally independent rollout rewards with pass rate $p = p_\theta(x)$ and $K = \sum_i r_i$. Let $s_i = \nabla_\theta \log \pi_\theta(y_i | x)$ be the response score and $\hat{g} = \sum_i w_i s_i$ the group update. Then pass@ $N = 1 - (1 - p)^N$. We analyze the zero-stabilizer idealization of the released practical centered MaxRL convention, which assigns

$$w_i = \frac{r_i}{K} - \frac{1}{N} \quad \text{when } K > 0, \quad w_i = 0 \quad \text{when } K = 0.$$

All-pass groups also have zero contrast. The released implementation uses a finite denominator stabilizer ε ; for $K > 0$ this multiplies the ideal coefficients by $K/(K + N\varepsilon)$. Thus the identities below are exact for the stated zero- ε convention rather than for its finite- ε magnitude.

Lemma 1 (Practical truncation order). The expectation of the drop- $K = 0$ practical estimator equals the gradient of the $(N - 1)$ -term truncated maximum-likelihood objective, not the N -term objective.

For $J_T(p) = -\sum_{t=1}^T (1 - p)^t / t$, direct conditioning gives

$$\mathbb{E}\hat{g} = \left(\frac{1 - (1 - p)^N}{p} - (1 - p)^{N-1} \right) \nabla p = \left(\sum_{j=0}^{N-2} (1 - p)^j \right) \nabla p = \nabla J_{N-1}.$$

The correction matters whenever a theoretical truncation parameter is mapped to a deployed group size. Details and the non-dropped control-variate convention are in App. A.

Proposition 1 (Exact coefficient mass). The expected absolute coefficient mass of the practical estimator is

$$A_N(p) = \mathbb{E} \left[\sum_{i=1}^N |w_i| \right] = 2\{1 - p - (1 - p)^N\} = 2(\text{pass@}N - \text{pass@}1).$$

For a mixed group with $0 < K < N$, successes and failures each contribute $(N - K)/N$ absolute mass. Therefore

$$A_N(p) = \frac{2}{N} \mathbb{E}[(N - K)\mathbf{1}\{K > 0\}] = 2\{(1 - p) - (1 - p)^N\}.$$

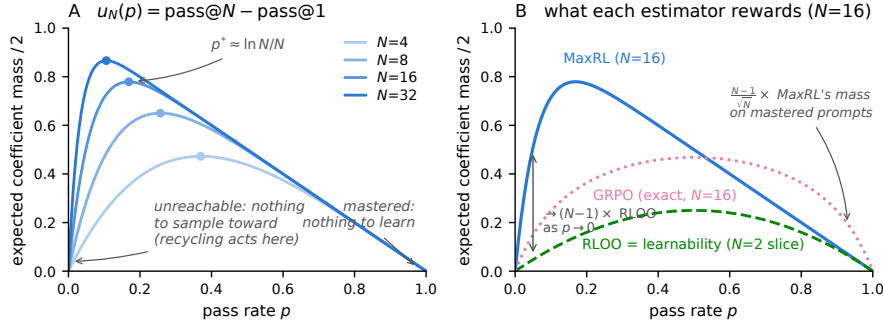


Figure 1: Estimator-conditioned activity. Left: $u_N = A_N/2 = \text{pass}@N - \text{pass}@1$ for several rollout counts; its peak is derived rather than tuned and moves toward harder tasks as N grows. Right: exact expected half-mass at $N = 16$ under zero-stabilizer MaxRL, standard gradient-averaged RLOO, and the zero-stabilizer sample-SD-normalized GRPO convention. Coefficient mass is sign-blind, so the curves motivate rather than prove the empirical coverage ordering.

This identity concerns coefficients, not gradient norm, signal-to-noise ratio, or expected improvement: score-gradient geometry can change its downstream effect.

The bridge to the expected update is nevertheless exact. With $\mu_+ = \mathbb{E}[s_i \mid r_i = 1, x]$ and $\mu_- = \mathbb{E}[s_i \mid r_i = 0, x]$,

$$\mathbb{E}[\hat{g} \mid x] = u_N(p)(\mu_+ - \mu_-), \quad u_N(p) = A_N(p)/2.$$

Thus u_N is the estimator-controlled scalar factor in the expected update; the conditional score geometry remains outside the mass calculation. It also has a direct event interpretation. For any designated rollout,

$$u_N(p) = \Pr(K > 0) - \Pr(r_1 = 1) = \Pr(r_1 = 0, K > 0) = (1 - p)\{1 - (1 - p)^{N-1}\}.$$

Thus activity is precisely the probability that one rollout fails while at least one peer in the purchased group succeeds. This makes the two dead zones transparent: an all-fail task supplies no positive contrast, while an all-success task supplies no negative contrast. It also clarifies why N is part of the curriculum definition rather than an implementation detail.

Consequences. The derivative of $u_N(p) = 1 - p - (1 - p)^N$ gives the unique peak

$$p_N^* = 1 - N^{-1/(N-1)} \simeq \frac{\log N}{N}.$$

Thus the relevant intermediate difficulty depends on rollout budget. At $N = 2$, $u_2 = p(1 - p)$, coinciding with a common learnability score. As $p \rightarrow 0$, MaxRL carries $(N - 1)$ times the coefficient mass of standard $1/N$ -averaged RLOO. As $p \rightarrow 1$, MaxRL and RLOO both decay at first order in $1 - p$, whereas sample-SD-normalized GRPO allocates relatively more mass to near-mastered prompts (App. A).

Fixed-completion check. We tested the arbitrary- N prediction on a 36-task CPU skill-chain testbed with $N \in \{2, 4, 8, 16, 32\}$, eight paired seeds, no recycling, and exactly 51,200 sampled completions per cell. Teacher decay was normalized per completion so changing N did not silently change its memory horizon.

The $N = 2$ identity reproduces bit-for-bit. For every $N > 2$, the rollout-aware utility beats the score-only $p(1 - p)$ comparator in all paired seeds. The mean gap increases across tested N , but only $5/8$ seed trajectories are monotone, so we do not claim a scaling law. Nor does u_N universally beat uniform: the mean contrast becomes positive only at $N \geq 8$. The paired positive-seed counts are $3/8, 3/8, 8/8, 7/8$, and $7/8$ as N increases. This is evidence for rollout awareness in one co-designed testbed, not a blanket curriculum advantage or an algorithm-level comparison with prior work.

Table 1: Post-guidance CPU check. Mean normalized pass-rate AUC differences at fixed sampled-completion budget. Seed signs are descriptive; the sweep was not pre-registered.

N	2	4	8	16	32
$u_N - p(1 - p)$	.0000	+.0307	+.0920	+.1526	+.1909
$u_N - \text{uniform}$	-.0106	-.0032	+.0309	+.0453	+.0836
wins vs. $p(1 - p)$	identical	8/8	8/8	8/8	8/8

Exact-probability counter-test. The stronger test does not generalize the sampler mapping. Under an internally hashed local protocol without an immutable public pre-execution commit, a fresh 24-block Digits contextual bandit computes the policy’s correct-class probability exactly before every update. Its internally frozen estimator-by-sampler interaction is $+.01589$ (95% CI $[-.01686, +.04712]$, exact sign-flip $p = .350$; 15/24 positive) and is not supported. MaxRL favors u_8 over $p(1 - p)$ by $+.20842$, but RLOO also favors u_8 , reversing its prediction; both estimator-matched samplers are below uniform. The activity algebra therefore generates testable hypotheses, not a universal learnability or curriculum-optimality law.

3 From mass to interventions

Task sampling. FrontierMax operationalizes the identity without changing the within-group estimator. Each task maintains discounted Beta evidence, draws a posterior pass rate $\tilde{p}_x$, and samples approximately as

$$q(x) = \frac{\epsilon}{M} + (1 - \epsilon) \frac{[u_N(\tilde{p}_x)]^\gamma}{\sum_{j=1}^M [u_N(\tilde{p}_j)]^\gamma}.$$

The floor ϵ , concentration γ , decay, and posterior estimator remain empirical choices; the theory contribution is the utility and its N -dependence, not a parameter-free scheduler.

The same utility also admits an exact deterministic posterior priority. If $p_x \mid \mathcal{D} \sim \text{Beta}(a_x, b_x)$ and $(z)_N$ denotes the rising factorial, then

$$\mathbb{E}[u_N(p_x) \mid \mathcal{D}] = 1 - \frac{(b_x)_N}{(a_x + b_x)_N} - \frac{a_x}{a_x + b_x}.$$

This is distinct from the Thompson-sampled rule above. Since u_N is strictly concave for $N > 1$, Jensen’s inequality gives $\mathbb{E}[u_N(p_x)] \leq u_N(\mathbb{E}[p_x])$: scoring only the posterior mean systematically overstates expected activity, while the analytic gap is the nonnegative, closed-form uncertainty penalty

$$u_N\left(\frac{a_x}{a_x + b_x}\right) - \mathbb{E}[u_N(p_x)] = \frac{(b_x)_N}{(a_x + b_x)_N} - \left(\frac{b_x}{a_x + b_x}\right)^N.$$

Which posterior treatment learns best remains an empirical question.

Contribution and evidence boundary. Coefficient activity is the shared quantity, not a license to conflate a proved identity, a tested fixed-pool teacher, and an untested replay overlay. Table 2 makes the paper’s evidence ladder explicit.

Predictions. The mass curves motivate, but do not prove, an empirical expectation: MaxRL and variance-normalized GRPO can exhibit different coverage orderings under a common sampler. This need not imply a sampler-by-estimator interaction and requires controlled measurement.

4 Evidence

The main evidence has two roles. The Acrobot tournament directly isolates the deployed- N score shape under a fixed scaffold, while the maze factorial tests an estimator-conditioned

Table 2: Contribution–evidence map. Status is attached to the specific object evaluated, not transferred across rows.

Object	Claim/mechanism	Evidence allowed here
Coefficient identity	Exact A_N , $T = N - 1$, and posterior-integrated u_N for the stated practical-MaxRL convention	Proved; estimator activity only, not learning progress
Direct teacher	fixed-pool Thompson-scored u_N with proportional sampling and a uniform floor	Scoped empirical; Acrobot supports the score in one fixed family, while Digits rejects a universal mapping
Coefficient-Activity PLR	Analytic posterior activity replaces MaxMC priority inside robust PLR; replay and staleness remain PLR mechanisms	Engineering only / hold; no AMaze, robotics, or minimax-baseline performance evidence

coverage ordering on real neural gradients. The preceding Digits counter-test supplies the negative boundary. The score’s positive support is deliberately small-scale and fully controlled; the neural maze family tests an estimator-side prediction, not the score.

4.1 Fresh fixed-pool Acrobot score test

We first isolate the arbitrary- N score on official Gymnasium Acrobot-v1. Eight nested tip-height predicates share one task-blind H64 actor (640 parameters); the hardest predicate is exactly native Acrobot success. All arms use practical MaxRL with $N = 16$, no hindsight, learning rate 3×10^{-4} , and a nominal two million paid environment transitions. Twenty fresh paired seeds compare uniform sampling, the common-scaffold $p(1 - p) = u_2$ score, and u_{16} under identical discounted Beta tracking. The internally frozen primary is target-uniform normalized transition-AUC — evaluation success averaged uniformly over the eight predicates, integrated over the paid-transition axis, and normalized by the budget — for $u_{16} - p(1 - p)$, with support requiring a positive exact two-sided sign-flip test at .05 and a point estimate of at least +.01.

Table 3: Fresh Acrobot tournament. Paired differences use 20 fresh seeds under the source/runtime-locked protocol. The two uniform comparisons form one Holm family.

contrast	mean Δ	95% boot. CI	exact p	Holm p	decision
$u_{16} - p(1 - p)$	+.0480	[+.0209,+.0738]	.0034	—	supported by rule
$p(1 - p)$ –uniform	-.0062	[-.0226,+.0122]	.5078	.5078	not supported
u_{16} –uniform	+.0419	[+.0218,+.0606]	.0008	.0016	supported

The arm means are .64523 (uniform), .63907 ($p(1 - p)$), and .68711 (u_{16}). The primary clears both frozen criteria; 15/20 paired differences are positive. The result therefore supports the deployed- N score shape against its $N = 2$ ablation in this fixed family, while the $p(1 - p)$ arm itself is not supported over uniform by the frozen secondary test.

The mean $u_{16} - p(1 - p)$ contrasts remain descriptively positive on sampled-group (+.05294) and optimizer-update (+.05366) axes. u_{16} also receives 2.40 fewer groups and 3.42 fewer updates per million transitions, so it did not win by receiving more updates; these non-confirmatory axes do not identify a causal mechanism. Native-success transition-AUC is .31085, .30521, and .37615 across the three arms. App. C reports the near-equal observed mass means between adaptive arms and records the audited but RNG-confounded precursor separately.

4.2 Externally recorded fresh-wave maze result

We next retain the maze factorial as a secondary estimator measurement; it does not test the acquisition score.

Design. The testbed is a 1.26M-parameter transformer trained on procedurally generated 17×17 mazes. Each balanced block shares a seed, warm start, task pool, and evaluation schedule across {MaxRL, GRPO} $\times$ {uniform, FrontierMax}. The outcome is time-integrated

pass@8 coverage relative to the warm-start checkpoint, not final reward, at identical deployed optimizer settings with no per-estimator learning-rate tuning.

Each run uses $N = 32$, eight prompt groups per training step, and 250 steps. Evaluation occurs every 25 steps on 16 held-out mazes per level; coverage AUC is the mean pass@8 over steps 25–250 minus the warm-start value.

The first factorial rejected its stronger externally recorded endpoint claim (3/6 uniform and 4/6 FrontierMax blocks with the predicted sign), and we retracted the earlier zero-exception cohort claim. Wave-1 time-integrated coverage was positive in all blocks under each sampler, but that outcome was discovered exploratorily; the external record specifies it as wave 2’s primary, with a falsification rule, before new seeds ran, though the locking commit object is not vendored in this checkout.

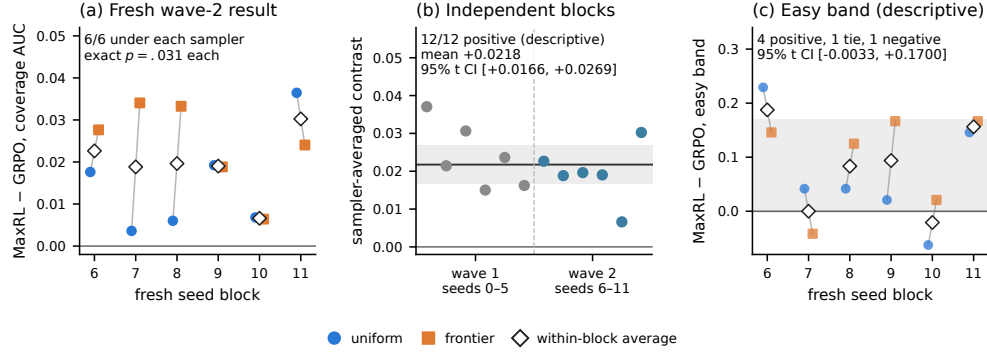


Figure 2: Maze factorial at the independent seed-block unit. (a) The externally recorded fresh-wave result is positive in 6/6 blocks under each sampler (exact sign $p = .03125$ each). Connected observations share a block. (b) Sampler-averaging gives wave-2 mean $+0.0195$ and post-hoc 95% CI $[+0.0115, +0.0275]$; 12/12 cross-wave block averages are descriptive. (c) Easy-band localization is suggestive: four positive blocks, one tie, one negative, and an interval containing zero.

Result. The wave-2 primary is positive in all six fresh blocks under uniform sampling and all six under FrontierMax (exact two-sided sign $p = .03125$ per sampler). Six blocks put this sign test at its granularity floor (6/6 is the only outcome clearing .05; the frozen 5/6 bar could not), so the inferential anchor is the block-averaged mean: averaging the correlated sampler observations within each block yields $+0.0195$, post-hoc 95% t interval $[+0.0115, +0.0275]$, with every leave-one-block-out interval above zero. Across both waves, all 12 block averages are positive, but that statement is descriptive because wave 1 selected the outcome.

The data support an estimator-conditioned coverage ordering at common deployed optimizer settings, not universal estimator superiority or the strong mechanism account we initially expected. The recorded pair-level easy-band criterion was met, yet block aggregation leaves four positive, one tied, and one negative easy-band contrast with an interval crossing zero. Localization therefore remains suggestive. We froze an AUC robustness multiverse, but the checkpoint trajectories required to evaluate its warmup, integration, and horizon variants remain external (App. C).

Secondary records. Paid-probe selection, failure recycling, and the historical Countdown aggregate are reported only in App. C. They support no named-method superiority claim. Their relevant measurement lesson is that raw per-task outcomes are needed to distinguish mean accuracy from standard pass@ k .

5 Related work

Our score is a distinct, testable functional form in a crowded neighborhood: derived from the deployed estimator’s own algebra, with a peak that migrates with the rollout budget, and used for task selection rather than objective redesign. Superiority over the methods below is untested and not claimed.

Concurrent exact analyses of group estimators. A concurrent cluster derives closed-form expected update magnitudes for group-relative estimators on the standard-deviation-normalized branch: the group-SD identity with DAPO’s silent-group mass $p^G + (1 - p)^G$ (Bay, 2026), and an expected absolute GRPO advantage of $2\sqrt{p(1 - p)}$ motivating 50%-success targets inside a learned-curator bandit (Gu et al., 2026); energy- and variance-based selection scores make the same alignment claim empirically (Cui, 2026). On that branch the per-rollout magnitude peaks at $p = .5$ independent of N , with N entering through the silent-group factor; the practical-MaxRL mass is instead the estimator’s own pass@ N gap, unnormalized, with a peak that migrates toward harder tasks as $\ln N/N$ — the functional form the fixed-completion sweep and the Acrobot tournament test. MoPPS (Qu et al., 2026) anticipates the posterior machinery (streaming Beta posteriors, Thompson-sampled pass rates, targeting $p \approx .5$); our novelty claim is only the utility sampled toward.

Curricula and rollout allocation. Intermediate-difficulty sampling predates this work. Under ProCuRL’s experimental assumption that the unknown optimal-policy success is one, its proximal score reduces to $p(1 - p)$ (Tzannetos et al., 2023), while SFL (Rutherford et al., 2024) and LILO (Foster et al., 2025) prioritize binary reward variance. Our distinction is convention-specific: $u_N = A_N/2$ is the arbitrary- N expected half-mass of the deployed centered MaxRL rule. VCRL (Jiang et al., 2025) thresholds realized group variance; PLR (Jiang et al., 2021) and ALP-GMM (Portelas et al., 2020) use TD error, staleness, or temporal learning progress, and can therefore react to forgetting where instantaneous coefficient mass cannot. Our AMaze implementation is explicitly PLR with a new replay priority, not a new replay architecture: robust replay gating, buffer admission, rank and staleness mixing, and with-replacement draws remain those of the pinned minimax substrate (Jiang et al., 2023). The narrow novelty is that the priority is the deployed-estimator- and rollout-conditioned posterior expectation above. Tie-aware ranking is shared with the matched MaxMC control and is treated as an experimental-control correction. This implementation is engineering-only: AMaze, robotics transfer, and performance against PLR, ACCEL, or other minimax baselines remain Hold and outside the evidence in this paper. Generator-based UED — PAIRED’s adversary (Dennis et al., 2020), ACCEL’s level editing (Parker-Holder et al., 2022) — creates tasks rather than pricing a fixed pool and has no native analogue in our Acrobot threshold family.

GRESO (Zheng et al., 2025) and DPS (Mao et al., 2026) select prompts before rollout from historical or predicted reward dynamics, and VIP (Nguyen et al., 2026) varies per-prompt rollout counts to cut predicted gradient variance — close operational baselines, but none uses the fixed- N half-mass of the deployed practical MaxRL convention.

Adaptive allocators vary per-prompt group size or stage extra rollouts toward high-value prompts (HORA (Wang et al., 2026), VIGOR (Jiang et al., 2026), DARS (Yang et al., 2025), Reinforce-Ada (Xiong et al., 2025)); FrontierMax instead selects the task distribution before generation at fixed N . BBG (Yuan et al., 2026) gates updates by posterior marginal pass@ k contribution, and boundary-aware Curriculum RL (Cai et al., 2026) adds external guidance near an estimated reasoning boundary. These criteria target hit probability, variance, exploration, boundary metrics, or signal creation—not practical-MaxRL coefficient mass. RL2ML (Zheng, 2026) separates finite-rollout objectives, update scale, variance, and metric gain; its $T = N$ alignment retains $K = 0$ via the direct estimator or a score-baseline control variate, while our $T = N - 1$ statement is for the deployed hybrid that centers only when $K > 0$ and zeros $K = 0$.

Coverage measurement. RLVR can improve pass@1 while reducing high- k coverage (Yue et al., 2025; Yuan et al., 2026; Zhou et al., 2026). Entropy collapse supplies complementary

evidence of reduced exploration (Cui et al., 2025); Pass@ k objectives address the metric directly (Chen et al., 2025). We study whether an external curriculum changes that coverage under two estimators. The secondary hindsight and relabeling record is confined to App. C; it is not evidence for the acquisition score.

6 Limitations

Coefficient mass ignores score-gradient direction, parameter sharing, optimizer state, task transfer, and non-i.i.d. rollouts. It is a sampler-facing surrogate, not a guarantee of loss reduction or metric improvement. The fixed-completion N -sweep is post-guidance (designed after the theoretical guidance existed, hence not preregistered) and uses one co-designed synthetic chain. The exact-probability Digits counter-test is a contextual bandit rather than trajectory RL, and its internally hashed local lock lacks an immutable public pre-execution commit establishing timing. The fresh V2 Acrobot score test uses one fixed eight-threshold family, one 640-parameter learner, and no held-out task distribution or native named-method implementation. Its 20-seed choice was inherited from a historical precursor rather than set by prospective power analysis; its exact paired test also requires sign exchangeability under the sharp null. No arm pairs a mismatched- N score with a fixed deployed N , so the tournament supports rollout-aware difficulty targeting rather than peak-location specificity of the deployed- N shape. The four evidence families froze their primaries independently at $\alpha = .05$; we claim no paper-level error rate. Retained evaluation records do not include per-episode trajectories. The maze result uses a small transformer, one task family, and common rather than estimator-tuned learning rates; its easy-band mechanism remains descriptive.

The Coefficient-Activity PLR row in Table 2 is an engineering specification, not evidence. No AMaze or robotics performance is reported, and no claim of outperforming PLR, robust PLR, ACCEL, or another minimax baseline is made; all remain Hold. The paid-probe, recycling, Countdown, gate, and transfer records are secondary and confined to App. C.

The external execution record describes the maze wave-2 outcome as specified before execution, but its locking commit object is not vendored in this checkout. The release can audit the stored protocol and verdict, not independently verify their timing. The Acrobot tournament lock and digest are vendored and internally bind source, runtime, gates, and artifacts, but this checkout likewise lacks an immutable public pre-execution commit establishing the study’s timing.

Artifact coverage remains incomplete: maze checkpoint trajectories are external, and the Digits full replay payload remains local without a download URI. App. D itemizes secondary-study gaps alongside the local 53-row compact registry and the distinct release-branch audit object. These gaps limit auditing and preclude a broad neural claim.

7 Conclusion

The estimator belongs in the evaluation of data-selection design: its coefficient algebra characterizes which tasks can emit activity at a chosen rollout count. A separately specified estimator-conditioned coverage ordering was observed under both tested samplers. In a fixed classical-control family, u_{16} scoring outperformed both u_2 scoring under the same deployed $N = 16$ estimator and uniform sampling; the Digits study shows that the estimator-to-sampler mapping is not universal. The maze result supplies supporting estimator-conditioned neural measurement, not direct evidence for the score or replay overlay. Data-selection methods should therefore be evaluated with the estimator they ship on, with pass@ k computed from retained task outcomes beside mean accuracy, and coefficient mass treated as a testable allocation hypothesis rather than a universal learning law.

Reproducibility statement

The repository includes exact-enumeration tests, a one-command artifact check, structured seed-block analyses, stored protocol and verdict documents, figure derivation scripts, a source-locked 60-run Acrobot tournament with deterministic retained-ledger reanalysis, and a generated 53-row compact registry for locally indexed maze, Countdown, and GSM8K artifacts. Secondary records and receipt manifests remain documented in the appendix. Registration timing objects that remain external are disclosed rather than treated as locally auditable. The independent unit is stated for every central result; known raw-data gaps and secondary-study provenance are disclosed in App. D.

AI use statement

AI assistants supported planning, implementation, debugging, literature discovery, statistical and artifact auditing, figure preparation, and manuscript editing. The authors selected the questions and decision rules, ran the studies, verified derivations, code, numerical results, and citations, and take responsibility for every claim.

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A Proof details

Truncation order. For $K > 0$, conditioning on one sampled response and using exchangeability gives

$$\mathbb{E}\hat{g} = \left(\frac{1 - (1-p)^N}{p} - (1-p)^{N-1} \right) \nabla p.$$

Since $1 - (1-p)^N = p \sum_{j=0}^{N-1} (1-p)^j$, subtracting the final term leaves $\sum_{j=0}^{N-2} (1-p)^j \nabla p = \nabla J_{N-1}$. If the centered control-variate term is retained on $K = 0$, its contribution is $(1-p)^{N-1} \nabla p$. The expectation becomes $\sum_{j=0}^{N-1} (1-p)^j \nabla p = \nabla J_N$. Thus the direct/non-dropped convention has order N , while the deployed hybrid convention above has order $N-1$; theorem and implementation must use the same convention. At audited MaxRL source commit 7197bbb, the released path computes $(r_i - \bar{r})/(\bar{r} + \epsilon)$ and zeros all-failure groups; after the deployed gradient average, its $\epsilon \rightarrow 0$ coefficients match this convention up to common scale. The parsed truncation-order setting is inactive on that path.

Coefficient mass. For $0 < K < N$, each success has coefficient $(N - K)/(KN)$ and each failure has coefficient $-1/N$. The two classes each contribute $(N - K)/N$ in absolute value. For $K \in \{0, N\}$ the deployed centered estimator contributes zero. With $K \sim \text{Bin}(N, p)$,

$$\begin{aligned} A_N(p) &= \frac{2}{N} \sum_{k=1}^{N-1} (N - k) \binom{N}{k} p^k (1 - p)^{N-k} \\ &= \frac{2}{N} \{\mathbb{E}[N - K] - N \Pr[K = 0]\} \\ &= 2\{1 - p - (1 - p)^N\}. \end{aligned}$$

The peak follows from $u'_N(p) = N(1 - p)^{N-1} - 1$ and strict concavity for $N > 1$.

Posterior-integrated activity. For $p \sim \text{Beta}(a, b)$, $\mathbb{E}[p] = a/(a + b)$ and the beta-function ratio gives

$$\mathbb{E}[(1 - p)^N] = \frac{B(a, b + N)}{B(a, b)} = \frac{(b)_N}{(a + b)_N}.$$

Substitution in $u_N(p) = 1 - (1 - p)^N - p$ proves the deterministic posterior priority in §3. Subtracting it from $u_N(\mathbb{E}[p])$ gives the printed nonnegative Jensen gap.

Estimator limits. Under the deployed $1/N$ gradient-average normalization, RLOO expected half-mass is $p(1 - p) = u_2(p)$, so $u_N(p)/u_2(p) \rightarrow N - 1$ as $p \rightarrow 0$. Near mastery, both are first order in $q = 1 - p$. With sample-SD normalization, GRPO group mass is proportional to $\mathbb{E}[\sqrt{K(N - K)}]$ up to deployed normalization constants; its relative tail scaling is verified by exact enumeration in `verify_guidance_math.json`. These are coefficient statements, not gradient-direction statements.

Dependent rollouts. The per-group identity does not require independence. For any binary group,

$$\mathbb{E} \sum_i |w_i| = 2\{\Pr(K \geq 1) - \mathbb{E}[K]/N\}.$$

With a common marginal p , the second term is p ; independence is used only to replace $\Pr(K \geq 1)$ by $1 - (1 - p)^N$. In a post-guidance exact beta-binomial CPU stress test, count-distribution enumeration matched the general identity to 2.58×10^{-13} . At $\rho = .10$, the activity peak moved from .169 to .234 for $N = 16$ and from .106 to .186 for $N = 32$. This dependence model shows how increasing positive correlation can lower activity inside that family; pairwise correlation alone does not determine a general joint law. The result is a scope diagnostic, not evidence that coefficient mass causes learning or that beta-binomial dependence describes neural rollouts.

B Algorithms and estimator contracts

Posterior teacher. Each task stores discounted successful and failed rollout counts. A Thompson draw $\tilde{p}_x$ is mapped through u_N and mixed with a uniform floor. Teacher state, sampler cursor, and checkpoints restore together. Observations use pre-relabel rewards so recycled outcomes do not inflate the requested task’s posterior.

Coefficient-Activity PLR (engineering only). The AMaze implementation is a narrow overlay on robust PLR in the pinned minimax substrate (Jiang et al., 2021; 2023). It uses cumulative Beta evidence per replay-buffer slot and replaces MaxMC priority with deterministic $\mathbb{E}[u_N(p) \mid \mathcal{D}]$ under exact $N = \text{n_eval}$ grouping. Buffer admission, the robust replay gate, inverse-rank mass, staleness mixing, and with-replacement draws remain PLR mechanisms. Tie-aware ranking is shared with the matched MaxMC control. This contract is engineering-only: no AMaze, robotics, or baseline-performance endpoint is reported or implied.

Relabeling (secondary). For an all-fail request, a verified achieved goal can create an auxiliary update where repeating the request would emit none. A parser proposes that goal from a failed response, and the environment’s exact verifier must accept the response under that goal. Prompt conditioning and explicit response goal declarations are rewritten; answer spans, arithmetic, decimals, signed values, and unknown contexts are preserved. The reported-run implementation is frozen, while a conservative release wrapper and 26 CPU regression/property tests protect future runs. Relabeled groups remain auxiliary selected-data updates; no importance ratio converts them into fresh on-policy target samples. The intervention changes both update dose and direction, and its achieved-goal distribution is biased toward outputs the policy already reaches (Andrychowicz et al., 2017; Ghosh et al., 2021).

C Experimental protocols and secondary results

Fixed-completion N -sweep. The environment contains three nested 12-level chains and ten actions. Every cell runs exactly 51,200 completions, with arms for uniform, $p(1 - p)$, and u_N sampling under the practical MaxRL estimator, no hindsight, and eight paired seeds. Checkpoints share a completion clock. The post-guidance analysis is descriptive.

Digits factorial (internally frozen negative). On a stored 1,077/360/360 train/development/test split of sklearn Digits, a 4,810-parameter float64 policy recomputes each example’s exact correct-class probability before every update. We crossed practical MaxRL and RLOO with uniform, $p(1 - p)$, and u_8 sampling. Five-rate development selected learning rate .1 for MaxRL, RLOO, and the common-rate check. Across 24 fresh paired blocks, the internally frozen interaction was +.01589 (paired bootstrap 95% CI $[-.01686, +.04712]$, exact two-sided sign-flip $p = .350$; 15/24 positive), so it was not supported despite passing the treatment-delivery gate. MaxRL did favor u_8 over $p(1 - p)$ by +.20842 (CI $[+.16791, +.24744]$; 23/24), but RLOO’s frozen prediction reversed: $p(1 - p) - u_8 = -.17665$ (CI $[-.22042, -.13593]$; 0/24). Moreover, the matched samplers were below uniform for both MaxRL ($-.11279$) and RLOO ($-.37581$). Because every selected rate equaled .1, the tuned and common schedules have byte-identical ledgers and checkpoints and identical scientific summary content after removing phase/authorization labels; they are identity checks, not independent replications. This exact- p contextual bandit therefore supports neither the planned interaction nor a universal curriculum advantage; it shows why expected coefficient mass must remain an estimator-activity proxy rather than a learning law. Its internally hashed local lock is not backed by an immutable public pre-execution commit that independently establishes timing.

Acrobot V3 (historical). The frozen control uses official Acrobot-v1, eight nested thresholds from -1.5 through native success at 1.0, one shared H64 actor, practical MaxRL, $N = 16$, and no hindsight. Uniform and u_{16} receive 20 paired seeds and a nominal two million actual transitions per run, completing the final group. Evaluation uses 32 shared trajectories total, reused across all eight predicates, at initialization and each 100,000 transitions with fixed common random numbers. The internally frozen primary is target-uniform normalized transition-AUC; group diagnostics and evaluation curves for all 40 runs are vendored. A post-hoc audit found cross-seed overlap between actor-initialization and action-sampling RNG roots, so the retained effect and mechanism audit are descriptive rather than clean confirmation. The mechanism audit is deterministic from those records.

Acrobot fixed-pool tournament V2. The replacement reruns uniform and u_{16} and adds $p(1 - p) = u_2$ on fresh logical seeds 20000–20019. All 60 source/runtime-locked runs completed after an outcome-blind development gate; globally disjoint RNG roots and per-run group-diagnostic and aggregate-checkpoint ledgers were retained. The primary $u_{16} - p(1 - p)$ transition-AUC difference is +.04803 (paired bootstrap 95% CI $[+.02094, +.07385]$, exact sign-flip $p = .003361$), clearing the predeclared +.01 estimate filter; Table 3 gives the Holm-controlled uniform comparisons. Observed mass means are close between adaptive arms (.7344 versus .7490 per group; 115.57 versus 115.90 per million transitions), both descriptive paired intervals include zero, and no equivalence test was frozen. The p -value tests zero

rather than the SESOI, and pairing does not guarantee sign exchangeability. The $p(1 - p)$ arm isolates only the score on our scaffold, not full ProCuRL or SFL.

Paid-probe ProCuRL selection semantics. This is a separate 80-pair family, not an extension of the V2 table. It keeps the same eight thresholds, H64 actor, practical MaxRL estimator, $N = 16$, and 3×10^{-4} learning rate. Its four arms are ordinary uniform; a uniform sham with paid probes; source-faithful ProCuRL selection softmax $\{20\hat{p}(1 - \hat{p})\}$; and continuous-range-matched u_{16} selection. Probed arms run 20 episodes per task initially and at every crossed 5,120-student-transition boundary; all student and probe transitions count toward the nominal two-million paid budget. Across seeds 21000–21079, the internally frozen fixed-paid-AUC contrast u_{16} –ProCuRL is $+0.004894$ (paired SD $.022139$, $t_{79} = 1.9773$, $p = .05149$; 20,000-resample CI $[+.000110, +.009727]$; one-million sign-flip $p = .05161$). It misses both the .02 SESOI and the .05 paired- t criterion. The five Holm-family contrasts are ProCuRL–sham $-.001704$ (adjusted $p = .4568$), u_{16} –sham $+0.003190$ (.4547), ProCuRL–ordinary $-.313774$ (3.47×10^{-66}), u_{16} –ordinary $-.308880$ (9.48×10^{-67}), and sham–ordinary $-.312070$ (4.06×10^{-68}). Mean fixed-paid AUCs are .33771 (ProCuRL), .33942 (sham), .65149 (ordinary), and .34261 (u_{16}).

The probed arms average 28.0–28.1 sweeps and 140.5–141.1k student transitions versus 2.003M for ordinary uniform; 93.19–93.21% of their paid transitions are probes. They average 15.4–19.8 optimizer updates versus 259.1 ordinary-uniform updates, with mean paid-budget overshoot 69,960–72,939 transitions versus 2,984. All 320 confirmation and 12 outcome-blind development runs reconcile exactly, all 11 development gates pass, and the common-random-number invariants replay.

An earlier outcome-blind development wave was invalidated before any gate or arm contrast because 73 entropy-sum replays differed by at most 7.28×10^{-12} . Its seeds were burned and its incident receipt is retained; the reported family uses fresh seeds and a replacement lock.

The external confirmatory raw is 1,374,886,097 bytes. Its SHA-256 boundary is:

```
raw: b1f8756c249effab8c77101c8bca73ddf708a5e143c18fe8742fd5712fdd7c12
lock: b7c7f76f6aaffa1fe65557717bfe545f2ec850495d370cd97fe72a8871fc8d0f
analysis: 2010e30b5b15a212e2d6bdfaecd43d243ae5f468a96be613cf59744b9bc2fb38
```

The source bundle carries a 320-record external-raw manifest: compact mode checks its small-artifact bindings, while full verification requires the raw file and replays locked validation. The local lock is mechanical provenance, not an immutable public pre-execution registration.

Post-guidance rollout-allocation factorial. We also compared fixed $N = 16$, published-HORA-style posterior hit allocation (Wang et al., 2026), and a mass-aware marginal that counts four already-spent probe rollouts. A 2-sampler by 3-allocator CPU factorial uses 16 paired seeds and 51,200 completions per cell. The mass-aware arm improves pass@8 AUC over HORA-style allocation by $+0.00676$ (11/16 positive pairs) and over fixed groups by $+0.02517$ (14/16). However, it reduces realized coefficient mass per completion by .000306 relative to HORA-style allocation in all 16 pairs, falsifying the frozen mechanism prediction. Its group sizes are also extreme (cell-average maximum about 76 versus 57 for HORA and 16 fixed). We therefore treat the result as exploratory motivation for posterior-calibration and allocation-cap ablations, not a fourth contribution.

A frozen follow-up crossed two samplers, the two adaptive scores, four caps, and three information sources over 16 paired seeds (50 cells; 800 runs). Restricting to the 32 deployable-information adaptive cells, all 32 adaptive-minus-fixed mean pass@8-AUC contrasts were positive, with 22 unadjusted descriptive Student- t intervals above zero; no multiplicity-adjusted inference is assigned. Yet realized coefficient mass was lower in 28/32 cells, with 24 intervals below zero and none above zero. The remaining 16 oracle-preupdate cells are diagnostic. Thus the expanded result reinforces the learning-mass mismatch rather than a mediation claim. The frozen engineering rule chose cap 32: averaged over samplers for the fresh-group/same-step cells, it reduced maximum group size by 58.07% with an AUC change of $-.00271$ versus uncapped. This is a point-estimate filter for a future pilot, not

an equivalence test. The source/runtime lock is internally hashed but lacks a public pre-execution commit establishing timing.

Maze factorials. Wave 1 has six shared seed/warm-start blocks and six arms per block: MaxRL and GRPO under uniform and FrontierMax sampling plus two diagnostic GRPO arms. Its recorded endpoint hypothesis failed. Wave 2 has six fresh blocks and the four core arms. The external execution record names time-integrated pass@8 relative to warm start primary, with at least 5/6 positive blocks required under each sampler. The block analyzer reports sampler-specific exact sign tests, sampler-averaged contrasts, post-hoc intervals, and leave-one-block-out checks. No pooled p -value treats sampler observations as independent.

AUC robustness status. The frozen multiverse covers integration rule, warmup inclusion, horizons, and leave-one-checkpoint-out variants, but all 24 wave-2 checkpoint JSONLs and warm starts remain external. Structured summaries recover the anchors (uniform 6/6, mean $+0.1496$; FrontierMax 6/6, $+0.02404$; sampler-averaged 6/6, $+0.01950$), but cannot identify those variants; the analyzer therefore exits explicitly blocked.

Countdown. We fine-tune SmolLM2-360M with $N = 16$. The v2 pool uses operand permutation, parenthesization, and exact division. Tier 1 has 128 evaluation tasks and 16 verifier-scored samples per task. B1 disables recycling; B2 admits exact-verifier achieved goals; the committed scoreboard contains three-seed aggregates. ARM B disables recycling and uses two optimizer epochs on every live group; its three individual endpoints are vendored. Headline endpoints are step-60 mean@16 and VERL bootstrap best@16. The latter uses 1,000 with-replacement resamples of size 16 per prompt and is not standard unbiased pass@16; missing per-task outcomes prevent recomputation. A stored task-identity summary reports zero tier-1 and tier-2 SFT overlap and 27 exposed tier-0 tasks; the source manifests needed to recompute those counts are absent. Tier 0 is excluded from absolute claims.

The reported recycling-package aggregate moves tier-1 mean@16 from $.278 \pm .066$ to $.324 \pm .014$ while bootstrap best@16 moves from $.541 \pm .024$ to $.492 \pm .013$ (mean $\pm$ sample SD across three seeds). Complete three-seed B1/B2 records are not vendored, so this supports neither a paired seed test nor a named-method effect claim. It is retained only as a measurement caution: mean accuracy alone cannot reveal whether coverage was spent to buy it.

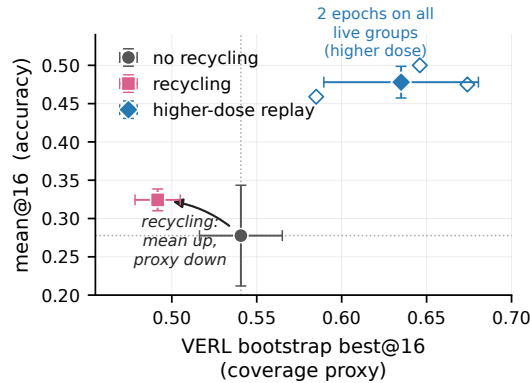


Figure 3: Secondary Countdown accuracy-coverage-proxy record. Error bars are sample standard deviations across three seeds; seed 3 for B1/B2 is aggregate-only. The recycling-package aggregate has higher mean@16 and lower VERL bootstrap best@16. The live-group replay control uses two optimizer epochs on every live group, so it cannot separate dose from update direction.

The externally specified replay control reaches mean@16 $.478 \pm .021$ and bootstrap best@16 $.635 \pm .046$, with all three endpoints vendored. Recycling permits at most 12 auxiliary

groups per 64 requested groups (18.75%; realized dose can be lower), whereas replay doubles optimizer epochs on every live group. This higher-dose alternative improves both logged metrics without identifying a dose-monotonic causal bound.

Negative branches. The artifact retains the failed maze endpoint, gate, Jugs, and robotics branches. GSM8K missed its delivery threshold (.601480 versus $< .60$) and is inconclusive; the single-seed Countdown per- k curve is exploratory. The corrected-code gate dose-response (frozen protocol, three paired seeds, independent A100 environment, retained 16-bit raw outcomes per task) passed its transfer gate — ungated recycling beat no-recycling on mean@16 in 3/3 seeds — and its manipulation check (rejection graded 0/.72-.80/.88-.90/.93 across gate strengths), but neither tested setting met the frozen usefulness criterion; per-seed signs were mixed at $n = 3$. Its raw outcomes also record, descriptively, one seed in which mean@16 rose while standard observed-set pass@16 fell $.656 \rightarrow .414$ — a non-proxy instance of the ambiguity the operational recommendation targets.

D Artifact and data accounting

bash reproduce.sh -build tests the identities and integrations, verifies declared manifest inputs and the local compact registry, runs stored endpoint spot checks, regenerates figures, and builds the canonical main_iclr2027.pdf plus the extended main.pdf. The compact registry has 53 records: 35 maze, 11 Countdown, and 7 GSM8K. The broader 562-row registry used during release-branch auditing is a different object; it is not copied over this checkout’s registry and is not used for the local count.

Missing external data are maze checkpoints; Countdown task manifests, outcomes, and B1/B2 seed 3, so the clean 101-task tier-0 analysis cannot be reconstructed; the 1.375 GB paid-probe confirmatory raw (its content hash and per-run receipt manifest ship); and the Digits 5.08 GB replay payload (block contrasts and a content manifest ship). Compact inputs and checksums are in paper/results/manifest.json and fail closed.